Types of Computers

Definition of a computer

- A computer is an electronic device that has storage, computations, input (data), output (data) and networking capabilities.
- The input and output data can be in different forms like text, images, audio and video.
- A computer processes the input according to the set of instructions provided to it by the user and gives the desired output.

Definition of a computer

- Computers are of various types and they can be categorized in two ways on the basis of size and on the basis of data handling capabilities.

Supercomputers

- Are the biggest and fastest computer (in terms of speed of processing data).
- Supercomputers are designed such that they can process a huge amount of data, like processing trillions of instructions or data just in a second.
- They have a thousands of interconnected processors in supercomputers.



Supercomputers Contd

- It is basically used in scientific and engineering applications such as weather forecasting, scientific simulations, and nuclear energy research.
- It was first developed by Roger Cray in 1976.
- Are used in the stock market or big organizations for managing the online currency world such as Bitcoin etc.
- Used in scientific research areas for analysing data obtained from exploring the solar system, satellites

Mainframe computers

- Mainframe computers are designed to support hundreds or thousands of users at the same time.
- Supports multiple programs simultaneously and can execute different processes simultaneously.
- All these features make the mainframe computer ideal for big organisations like banking, telecom sectors, etc., which process a high volume of data in general.

Mainframe computers. Contd

- It has high storage capacity and great performance.
- It runs smoothly for a long time and has a long life.

Minicomputer

- Minicomputer is a medium size multiprocessing computer.
- In this type of computer, there are two or more processors, and it supports 4 to 200 users at one time.
- Minicomputer is similar to Microcontroller.
- Minicomputers are used in places like institutes or departments for different work like billing, accounting, inventory management, etc.

Minicomputer. Contd

- It is smaller than a mainframe computer but larger in comparison to the microcomputer.
- Its weight is low.
- It is easy to carry anywhere.
- Less expensive than a mainframe computer.
- It is fast.

Workstation Computer

- A workstation computer is designed for technical or scientific applications.
- It consists of a fast microprocessor, with a large amount of RAM and a high-speed graphic adapter.
- It is a single-user computer.
- It is generally used to perform a specific task with great accuracy.

Workstation Computer. Contd

- It is expensive or high in cost.
- They are exclusively made for complex work purposes.
- It provides large storage capacity, better graphics, and a more powerful CPU when compared to a PC.
- It is also used to handle animation, data analysis, CAD, audio and video creation, and editing.

Personal Computer

- Personal Computers is also known as a microcomputer.
- It is basically a general-purpose computer designed for individual use.
- It consists of a microprocessor as a central processing unit(CPU), memory, input unit, and output unit.



Personal Computer. Contd

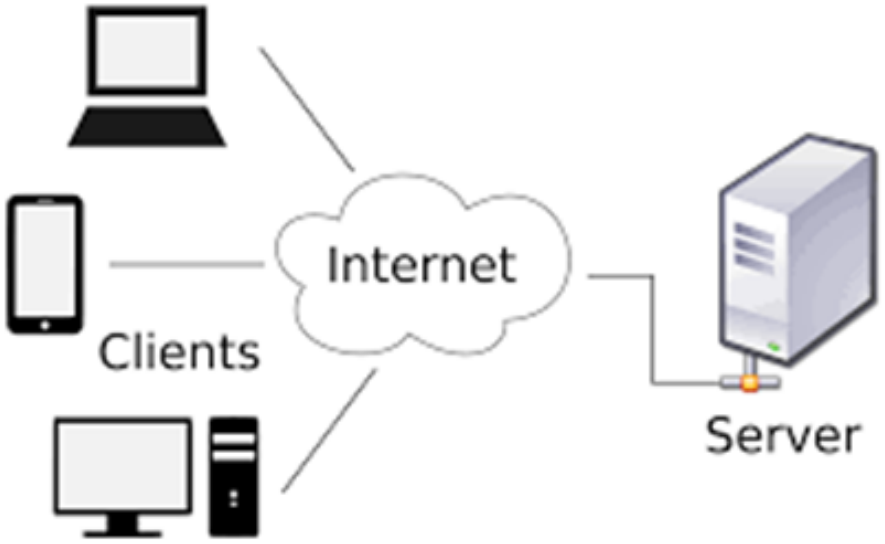
- This kind of computer is suitable for personal work such as making an assignment, watching a movie, or at the office for office work, etc. For example, Laptops and desktop computers.

Personal Computer. Contd

- In this computer, limited number of software can be used.
- It is the smallest in size.
- It is easy to use.

Server Computer

- Server Computers are computers that are combined data and programs.
- Electronic data and applications are stored and shared in the server computer.
- The working of a server computer is that it does not solve a bigger problem like a supercomputer but it solves many smaller similar ones.



Server Computer. Contd

- Examples of server computer are like Wikipedia, as when users put a request for any page, it finds what the user is looking for and sends it to the user.

Key Features of Server

- Servers come packed with several key features that make them indispensable in the tech world. Let's understand these features:

Key Features of Server

- **Scalability**: Refers to a server's ability to handle increasing workloads by adding resources. For instance, as your business grows, you can add more storage or processing power to your server without disrupting its operations.
- **High Processing Power:** Servers are equipped with powerful processors that can handle complex computations and manage large amounts of data, ensuring smooth and efficient operations.

Key Features of Server

- **Reliability: Servers are d**esigned to run continuously without failure, and often come with redundancy features, like backup power supplies and hard drives. This will ensure they stay up and running, even in the event of a hardware failure.
- **Better Collaboration:** Providing a centralized platform where users can pool resources and collaborate efficiently, servers make teamwork easier, irrespective of geographical locations.

Key Features of Server

- **Cost Efficiency:** While servers might seem expensive initially, they can save money in the long run by centralizing resources. This reduces the need for multiple devices and systems, leading to lower maintenance and hardware costs.
- **Security from Cyberattacks:** Equipped with advanced security features, secure servers provide protection against cyber threats. These include firewalls, intrusion detection systems, and encryption tools that ensure the safety of your data.

Analog Computer

- Analog Computer: Are particularly designed to process analog data.
- Analog data is continuous data that changes continuously and cannot have discrete values.
- An analog computer is used where we do not need exact values or need approximate values such as speed, temperature, pressure, etc.

Analog Computer. Contd

- It can directly accept the data from the measuring device without first converting it into numbers and codes.
- It measures the continuous changes in physical quantity.
- It gives output as a reading on a dial or scale. For example, speedometer, mercury thermometer, etc.

Digital Computer

- Digital computers are designed in such a way that they can easily perform calculations and logical operations at high speed.
- It takes raw data as input and processes it with programs stored in its memory to produce the final output.
- It only understands the binary input 0 and 1, so the raw input data is converted to 0 and 1 by the computer and then it is processed by the computer to produce the result or final output.

Digital Computer. Contd

- All modern computers, like laptops, desktops including smartphones are digital computers.

Hybrid Computer

- Hybrid means made by combining two different things. The hybrid computer is a combination of both analog and digital computers.
- Hybrid computers are fast like analog computers and have memory and accuracy like digital computers. It has the ability to process both continuous and discrete data.

Hybrid Computer. Contd

- It accepts analog signals as input then it converts them into digital form before processing the input data.
- A hybrid computer is widely used in specialised applications where both analog and digital data are required to be processed.



Hybrid Computer. Contd

- A processor which is used in petrol pumps that converts the measurements of fuel flow into quantity and price is an example of a hybrid computer.

Tablets and Smartphones

- Tablets and Smartphones are the types of computers that are pocket friendly and easy to carry.
- This is one of the best use of modern technology and have better hardware capabilities, extensive operating systems, and better multimedia functionality.
- They contain a number of sensors and are also able to provide wireless communication protocols.